LORDS INSTITUTE OF ENGINEERING AND TECHNOLOGY

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Department of Computer Science and Engineering (Data Science)

MAJOR PROJECT PHASE -I ABSTRACT SUBMISSION FORM (2026–27)

Semester: VII **Section:** B

Project Group ID : CSDMP230031

CULTURAL HERITAGE EDUCATION IN THE METAVERSE

Abstract:

The rapid advancement of immersive technologies such as the Metaverse, Virtual Reality (VR), and 3D visualization is transforming modern education by providing interactive and engaging learning experiences. Traditional cultural heritage education often relies on textbooks, static images, and conventional classroom methods, which may limit learners' ability to explore historical sites, artifacts, and cultural traditions effectively. To address these challenges, this project proposes GUYUK: Cultural Heritage Education in the Metaverse, an immersive educational platform that integrates 3D virtual environments, interactive cultural content, game-based learning, and storytelling into a unified learning experience. The platform enables users to explore digitally reconstructed historical sites and museums while accessing informative and educational content. It incorporates quizzes, exploratory tasks, and narrative-driven challenges to improve engagement, knowledge acquisition, and active participation. By combining immersive visualization with gaming principles, GUYUK transforms conventional heritage education into an interactive and enjoyable learning experience. The project contributes to **SDG 4**: Quality Education by promoting innovative learning approaches and **SDG 11**: Sustainable Cities and Communities by encouraging awareness and digital preservation of cultural heritage. Overall, GUYUK provides an engaging virtual platform for learning, exploring, and appreciating cultural heritage.

Keywords: Generative AI, Large Language Models, Retrieval-Augmented Generation, Business Intelligence, Sales Analytics, Marketing Analytics, Customer Behavior Prediction, Semantic Search, AI Assistant

Supervisor Approval

I hereby approve the above abstract for Major Project development.

Name of the Supervisor: Mr. Syed Mustaq Ali

Signature: _____ **Date**: _____

Coordinator **Head of the Department**